Nour Karawani

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Objective

Chemical Engineering graduate with strong teamwork, creative problem-solving, and analytical skills. Known for a fast learning curve and the ability to perform under pressure. Seeking a challenging role in chemical engineering that enables growth and meaningful contributions in process design, R&D;, or integration engineering.

Experience

R&D; Algorithm Developer & Data Analyst — Atiko Labs, Israel | 2023 – 2025
• Developed and optimized signal processing, statistical data analysis, and pattern recognition algorithms to enhance Surface-Enhanced Raman Spectroscopy (SERS) analysis.
• Improved signal-to-noise ratio (<2%) and enabled trace-level detection down to 1 µM in complex media.
• Delivered real-time, on-site diagnostics for non-invasive early detection, providing quick, reliable, and cost-effective solutions.
• Integrated analytical tools into customer-facing diagnostic platforms, improving accessibility and accuracy.

Industry 4.0 Automation Trainee — Moona – A Space for Change, Israel | 2024 – 2025
• Designed and deployed an automated box-sorting system integrating PLC programming, Yaskawa robotics, laser sensors, and Cognex cameras for precise and efficient sorting.
• Optimized pneumatic motion control and robotic process automation (RPA) to enhance operational speed, reduce energy consumption, and minimize manual labor.
• Collaborated with a multidisciplinary team to integrate and prototype system components, enhancing system reliability and adaptability.

Research Assistant – Water Treatment — Jacob Blaustein Institutes for Desert Research (BIDR), Israel | 2022 – 2023
• Conducted column experiments with zeolite and synthetic membranes for ammonium removal from wastewater.
• Optimized parameters for adsorption efficiency and membrane stability.
• Supervisor: Prof. Oded Nir.

Engineering Project – Nickel Recovery from Ni-MH Batteries — Ben-Gurion University, Israel | 2023 – 2024
• Collaborated in a team to design, control, and optimize the nickel recovery process using sulfuric acid (H_2SO_4) leaching, ensuring process efficiency, safety, and risk mitigation.
• Supervisor: Mr. Ronen Berman, RTA Engineering LTD.

Education

B.Sc. Chemical Engineering — Ben-Gurion University of the Negev, 2024 | GPA: 80.65
Relevant Courses: Structure & Properties of Semiconductors (90), Nano- & Molecularly Structured Surfaces (97), Microelectronic Devices (91).
AI Developer Program – Computer & Data Science — Ecom School, Expected Dec 2025

Publication

• Karawani, N., Bashouti, M.Y. (Under Review). *RamanSense: An Adaptive Algorithm for Enhanced SERS Analysis.*

Key Skills

• **Process & R&D;:** PFD/P&ID; design, mass & energy balances, chemical separation processes, experimental planning.
• **Programming & Data:** Python, SQL, MATLAB, Excel.
• **Automation & Control:** PLCs (Siemens, Unitronics, Allen-Bradley), Ladder Logic, Cognex cameras, Yaskawa/UR robotics, IIoT, Arduino.
• **Engineering Tools:** SolidWorks, ChemCAD, Visio, 3D printing.
• **Languages:** English (Advanced), Hebrew (Fluent), Arabic (Native).

Volunteering

Perach Tutor (2022–2024) and Edmond de Rothschild Ambassador (2024) — supported youth education, coordinated events, and gained leadership and teamwork experience.

References

• Mr. Ronen Berman – Engineering Project Supervisor, RTA Engineering LTD
• Prof. Oded Nir – Water Treatment Supervisor, BIDR
• Prof. Muhammad Y. Bashouti – R&D; Supervisor, Atiko Labs